

3 THE MOMENT-AREA METHOD

3.1 INTRODUCTION

The **Moment – Area method** was developed by Charles Ezra **Greene (1897)** based on **Mohr's theorems (1870)**, which relate the **geometry** of the deflected shape to the **curvature** of the beam.

3.2 MOHR'S 1ST THEOREM

Using the expression $d^2y/dx^2 = -M/EI$ and noting that $\theta = dy/dx$ we can express:

$$-d\theta/dx = M/EI \Rightarrow -d\theta = (M/EI) dx$$

Integrating both sides gives:

$$\begin{aligned} -\int_A^B d\theta &= \int_A^B (M/EI) dx \Rightarrow -(\theta_B - \theta_A) = \int_A^B (M/EI) dx \\ \Rightarrow \theta_A - \theta_B &= \int_A^B (M/EI) dx \quad \text{----- Eqn 1} \end{aligned}$$

Mohr's 1st theorem states that the change in slope due to flexure between any two points, say A and B, is equal to the area under the M/EI diagram between the two points.

3.3 MOHR'S 2ND THEOREM

The expression $d^2y/dx^2 = -M/EI$ may be multiplied by x on both sides and then integrated to obtain: -

$$-\int_A^B x (d^2y/dx^2) dx = \int_A^B x (M/EI) dx \quad \text{----- (i)}$$

v du 1st moment of area of the M/EI diagram between A and B, taken about A

The R.H. side of the above expression represents the 1st moment of area of the M/EI diagram between A and B, taken about A. The L.H. side may be treated as the product of the function v and the differential du .

In calculus, $\int v du = uv - \int u dv$, and therefore the L.H. side may be written as:

$$-\int_A^B x (d^2y/dx^2) dx = - \left\{ \left[x (dy/dx) \right]_A^B - \int_A^B (dy/dx) dx \right\} = - \{ (x_B \theta_B - x_A \theta_A) - (y_B - y_A) \}$$

If we set our reference axis to originate from A, the $x_A = y_A = 0$, and

$$-\int_A^B x (d^2y/dx^2) dx = -x_B \theta_B + y_B = y_B - x_B \theta_B$$

Hence eqn (i) becomes

$$y_B - x_B \theta_B = \int_A^B (M/EI) x dx \quad \text{----- (ii)}$$

The quantity $y_B - x_B \theta_B$ is the vertical deviation of the point A from a tangent drawn at the point B, and is denoted by $\Delta_{A/B}$. Eqn (ii) is therefore written as

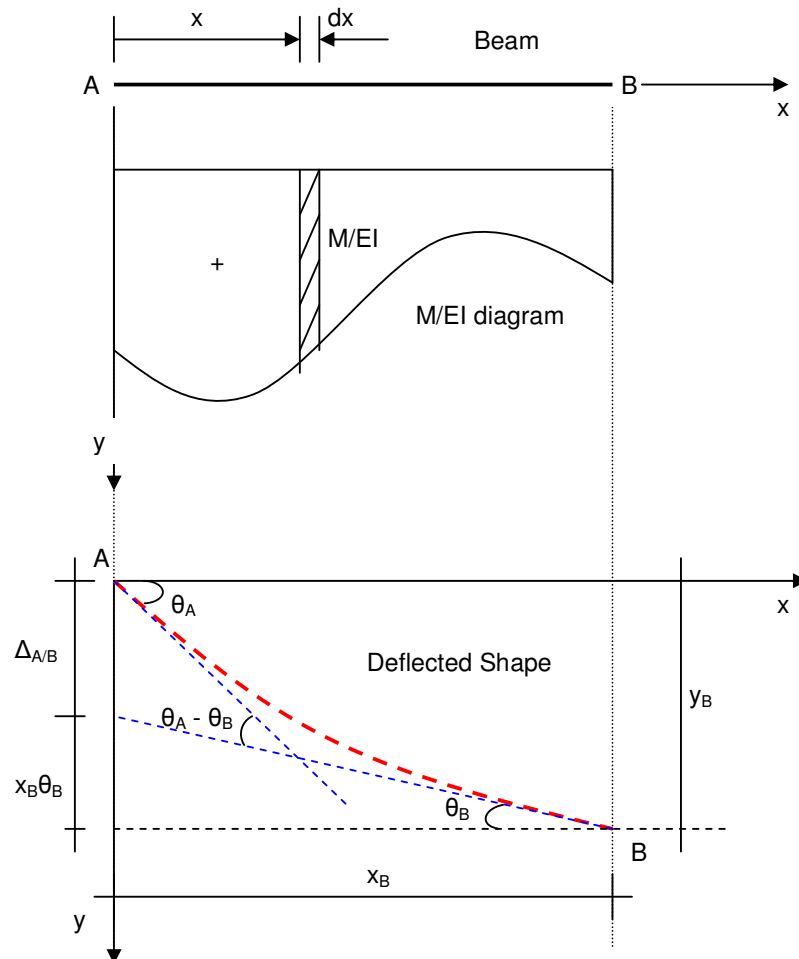
$$\Delta_{A/B} = \int_A^B (M/EI) x dx \quad \text{----- Eqn 2}$$

Mohr's 2nd theorem states that the vertical deviation of point A from the tangent drawn at point B is equal to the 1st moment of area of the M/EI diagram between A and B, taken about A.

[The deviation is measured positive upwards from the tangent drawn at B].

3.4 INTERPRETATION OF MOHR'S THEOREMS

Mohr's theorems may be interpreted diagrammatically as shown below.



3.5 SOLUTION PROCEDURE

The solution procedure may be outlined as follows: -

1. Analyse the beam (according to FCE 201) to obtain the **M/EI diagram**.
2. Sketch the **D-shape**.
3. Use 1st theorem to calculate slopes
[One of the slopes must be known]
4. Use 2nd theorem to calculate deflections
[Define the point of reference and its tangent carefully]

[Steps 3 and 4 may be interchanged].

Example 1

Determine θ_B and δ_B for the beam shown.

Assume $EI = \text{constant}$.

Solution

Using 1st theorem

$$\theta_A - \theta_B = \frac{1}{2} (L) (-PL/EI)$$

But $\theta_A = 0$ therefore

$$-\theta_B = \frac{1}{2} (L) (-PL/EI)$$

$$\theta_B = PL^2/2EI$$

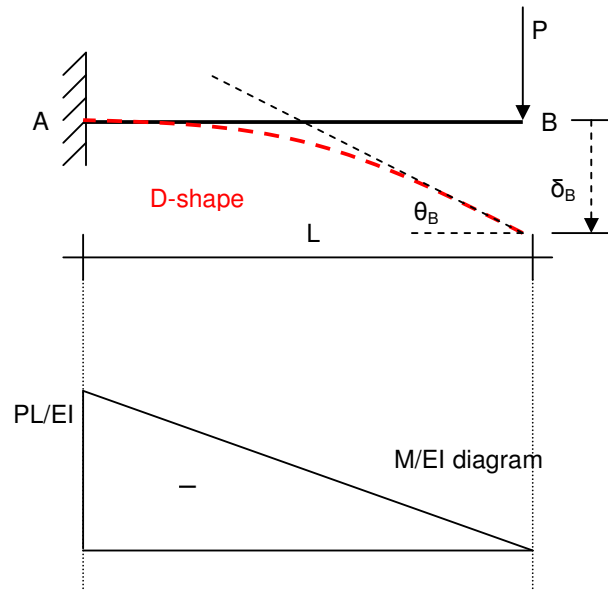
Using 2nd theorem

$$-\Delta_{B/A} = \frac{1}{2} (L) (-PL/EI) \left(\frac{2}{3}\right) (L)$$

$$\Delta_{B/A} = PL^3/3EI$$

By geometry $\delta_B = \Delta_{B/A}$ therefore

$$\delta_B = PL^3/3EI$$



Example 2

Determine θ_B and δ_B for the beam shown.

Solution

Using 1st theorem

$$\theta_A - \theta_B = \frac{1}{2} (L) (-PL/2EI) + \frac{1}{2} (L/2) (-PL/4EI)$$

But $\theta_A = 0$ therefore

$$-\theta_B = \frac{1}{2} (L) (-PL/2EI) + \frac{1}{2} (L/2) (-PL/4EI)$$

$$\theta_B = 5PL^2/16EI$$

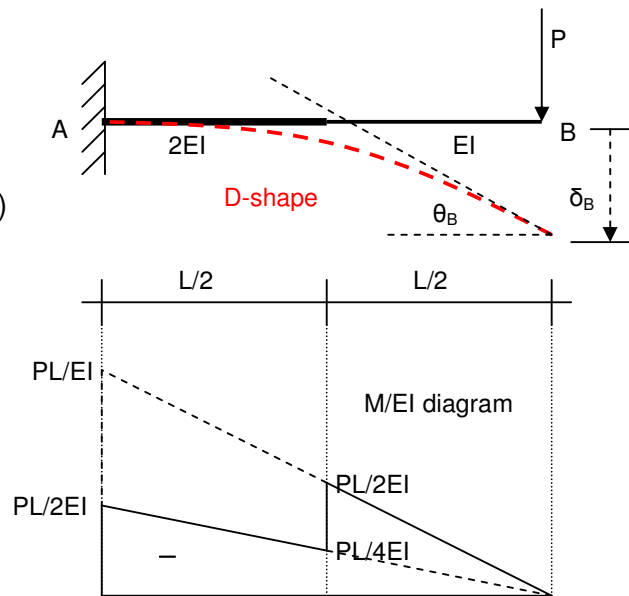
Using 2nd theorem

$$-\Delta_{B/A} = \frac{1}{2} (L) (-PL/2EI) \left(\frac{2}{3}\right) (L) + \frac{1}{2} (L/2) (-PL/4EI) \left(\frac{2}{3}\right) (L/2)$$

$$\Delta_{B/A} = 3PL^3/16EI$$

By geometry $\delta_B = \Delta_{B/A}$ therefore

$$\delta_B = 3PL^3/16EI$$



Example 3

Determine θ_B and δ_B for the beam shown.

Assume $EI = \text{constant}$.

Solution

Using 1st theorem

$$\theta_A - \theta_B = \frac{1}{3}(L) \left(-\frac{\omega L^2}{2EI} \right)$$

But $\theta_A = 0$ therefore

$$-\theta_B = \frac{1}{3}(L) \left(-\frac{\omega L^2}{2EI} \right)$$

$$\theta_B = \frac{\omega L^3}{6EI}$$

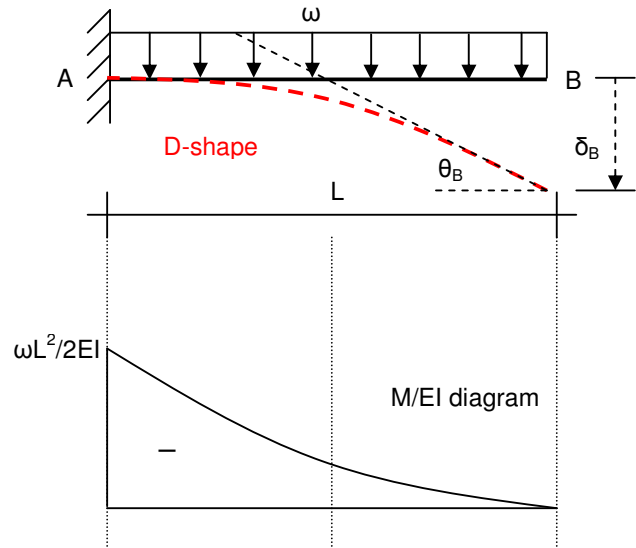
Using 2nd theorem

$$-\Delta_{B/A} = \frac{1}{3}(L) \left(-\frac{\omega L^2}{2EI} \right) \frac{3}{4}(L)$$

$$\Delta_{B/A} = \frac{\omega L^4}{8EI}$$

By geometry $\delta_B = \Delta_{B/A}$ therefore

$$\delta_B = \frac{\omega L^4}{8EI}$$



Example 4

Determine θ_B and δ_B for the beam shown.

Solution

Using 1st theorem

$$\theta_A - \theta_B = \frac{1}{3}(L) \left(-\frac{\omega L^2}{4EI} \right) + \frac{1}{3}(L/2) \left(-\frac{\omega L^2}{16EI} \right)$$

But $\theta_A = 0$ therefore

$$-\theta_B = \frac{1}{3}(L) \left(-\frac{\omega L^2}{4EI} \right) + \frac{1}{3}(L/2) \left(-\frac{\omega L^2}{16EI} \right)$$

$$\theta_B = \frac{3\omega L^3}{32EI}$$

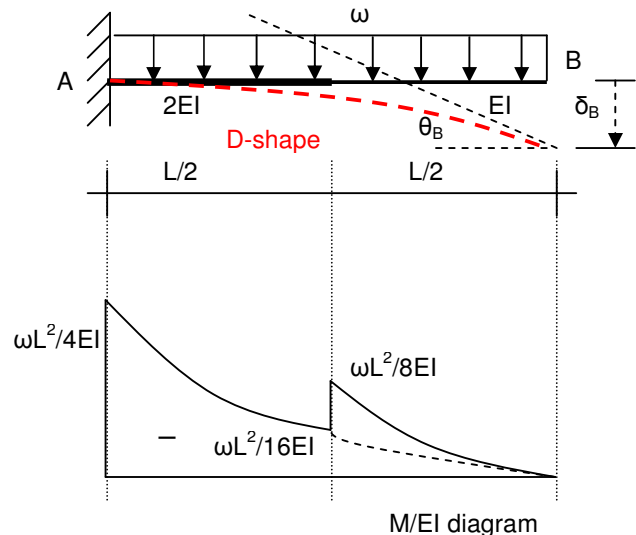
Using 2nd theorem

$$-\Delta_{B/A} = \frac{1}{3}L \left(-\frac{\omega L^2}{4EI} \right) \frac{3}{4}L + \frac{1}{3}(L/2) \left(-\frac{\omega L^2}{16EI} \right) \frac{3}{4}(L/2)$$

$$\Delta_{B/A} = \frac{17\omega L^4}{256EI}$$

By geometry $\delta_B = \Delta_{B/A}$ therefore

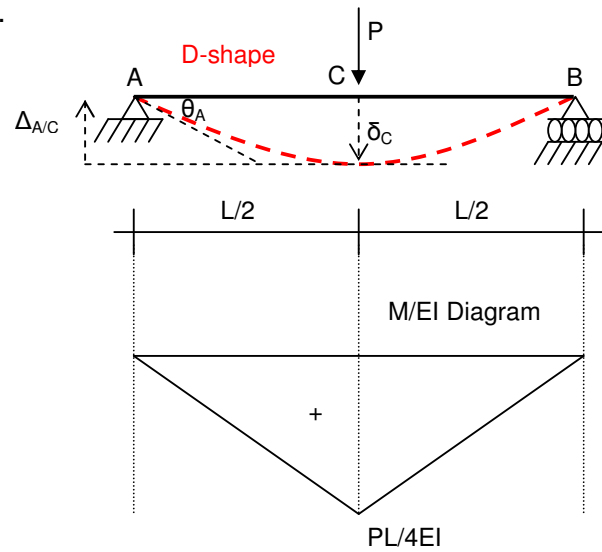
$$\delta_B = \frac{17\omega L^4}{256EI}$$



Example 5

Determine θ_A and δ_C for the beam shown.

Assume $EI = \text{constant}$.



Solution

Using 1st theorem

$$\theta_A - \theta_C = \frac{1}{2} (L/2) PL/4EI$$

But $\theta_C = 0$ therefore

$$\theta_A = \frac{1}{2} (L/2) PL/4EI \\ = PL^2/16EI$$

Using 2nd theorem

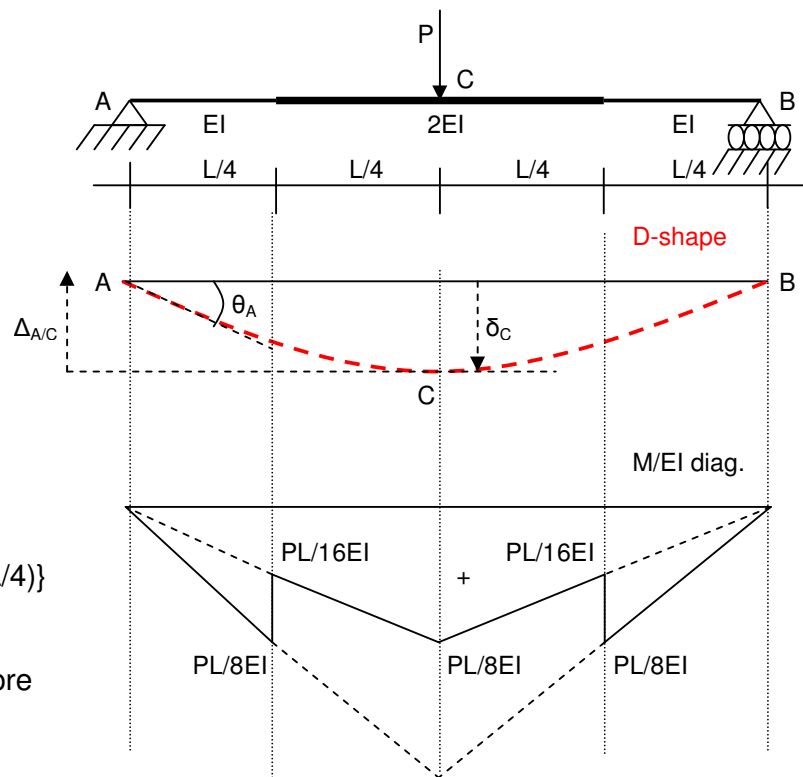
$$\Delta_{A/C} = \frac{1}{2} (L/2) PL/4EI \cdot \frac{2}{3} (L/2) = PL^3/48EI$$

By geometry $\delta_C = \Delta_{A/C}$ therefore

$$\delta_C = PL^3/48EI$$

Example 6

Determine θ_A and δ_C for the beam shown.



Solution

Using 1st theorem

$$\theta_A - \theta_C = \frac{1}{2} (L/2) (PL/8EI) \\ + \frac{1}{2} (L/4) (PL/16EI)$$

$$\theta_A = 5PL^2/128EI$$

Using 2nd theorem

$$\Delta_{A/C} = \frac{1}{2} (L/2) (PL/8EI) \left(\frac{2}{3} \right) (L/2) \\ + \frac{1}{2} (L/4) (PL/16EI) \left\{ \frac{2}{3} (L/4) \right\} \\ = 3PL^3/256EI$$

By geometry $\delta_C = \Delta_{A/C}$ therefore

$$\delta_C = 3PL^3/256EI$$

Example 7

Determine θ_A and δ_C for the beam shown.

Assume $EI = \text{constant}$.

Solution

Using 1st theorem

$$\theta_A - \theta_C = \frac{2}{3} (L/2) \omega L^2 / 8EI$$

But $\theta_C = 0$ therefore

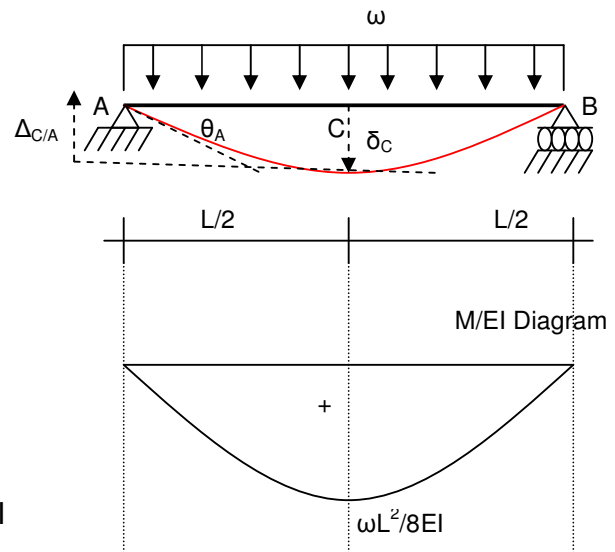
$$\theta_A = \frac{2}{3} (L/2) \omega L^2 / 8EI \\ = \omega L^3 / 24EI$$

Using 2nd theorem

$$\Delta_{A/C} = \frac{2}{3} (L/2) (\omega L^2 / 8EI) \frac{5}{8} (L/2) = 5\omega L^4 / 384EI$$

By geometry $\delta_C = \Delta_{A/C}$ therefore

$$\delta_C = 5\omega L^4 / 384EI$$



Example 8

Determine θ_A and δ_C for the beam shown.

Solution

[NB: Upper $[M/EI]$ is NOT standard]

$$\text{Upper } M/EI = (1/2)(\omega Lx/2EI - \omega x^2/2EI)$$

$$\text{Area} = A = \int_0^{L/4} (\omega Lx/4EI - \omega x^2/4EI) dx \\ \Rightarrow A = 5\omega L^3 / 768EI$$

$$A \bar{x} = (1/EI) \int_0^{L/4} (\omega Lx^2/4 - \omega x^3/4) dx \\ = 13\omega L^4 / 12288EI \\ \Rightarrow \bar{x} = 13L/80 \text{ from A}$$

Using Mohr's 2nd theorem

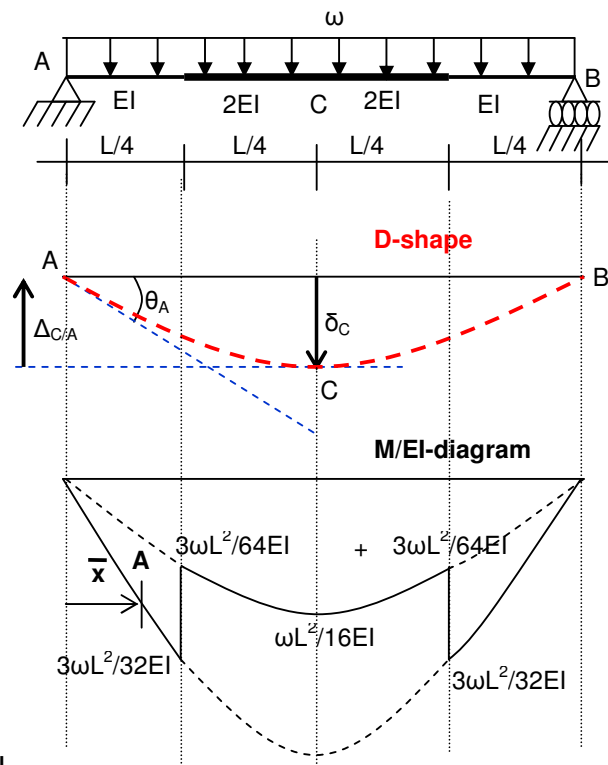
$$\Delta_{A/C} = (2/3)(L/2)(\omega L^2/16EI)(5/8)(L/2) \\ + (5\omega L^3/768EI)(13L/80) \\ = 31\omega L^4 / 4096EI$$

By geometry: $\delta_C = \Delta_{A/C}$

$$\Rightarrow \delta_C = 31\omega L^4 / 4096EI$$

Using 1st theorem

$$\theta_A - \theta_C = (2/3)(L/2)(\omega L^2/16EI) + 5\omega L^3/768EI \\ \Rightarrow \theta_A = 7\omega L^3 / 256EI$$



Thankx
Greene
for
Excellent
technique
